Axel Pilling

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EDUCATION

University of Arizona, W.A. Franke Honors College — Tucson, AZ

B.S. Chemical Engineering, Mathematics minor, accelerated three-year track; planned Accelerated M.S. (May 2029) — *May 2028*

- GPA 4.0/4.0 | National Merit Finalist | Coursework: Chemical Engineering Modeling, Mass Transfer, Physical Chemistry, Unit Operations Lab, Engineering Programming (MATLAB, Python), Differential Equations, Linear Algebra, Vector Calculus

EXPERIENCE

Simple Industry — Tucson, AZ

Co-Founder and Product Lead, operations software for small manufacturers — *May 2026 – Present*

- Lead product for software that runs production, inventory, BOMs, purchasing, and lot traceability at small fab shops: 3 shops in development and weekly feedback, 3 more in design. Coordinate a 3-person founding team.
- Run user research on the shop floor, return in 2 to 3 weeks with a clickable prototype, then ship and iterate on weekly working sessions with each shop.
- Replaced a job-ticket workflow operators were skipping with one “log what I did” action that updates inventory and traceability.
- Modeled raw material use the way the floor measures it (parts per sheet, cut length per part), captured at a part’s first run.
- Found a wedge when 3+ aerospace shops independently asked for AS9100/ITAR paperwork tracking; built material lots tied to mill test reports, certs, POs, and travelers.
- Build with AI coding agents (Claude Code, multi-agent pipelines) on TypeScript/Postgres; 4,000+ tests, weekly releases.

Freeport-McMoRan — Phoenix, AZ

Process Automation Engineering Intern — *May 2026 – Aug 2026*

- Audited about 3,000 control loops at all 11 sites via the PlantESP API; found about 1,000 misconfigured for two years.
- Defined a 100%-precision screening rule (OCE above 50, uptime above 75%) that replaced manual triage.
- Built a Python bulk-import pipeline and weekly audit the team still runs, plus an AI copilot for fixing loop configuration.
- Trained about 10 engineers and managers on AI tools; presented findings to technical leads, VPs, and C-suite.

Blowers Computational Chemistry Group, University of Arizona — Tucson, AZ

Undergraduate Researcher, computational quantum chemistry — *Sept 2025 – Present*

- Compute ideal-gas heat capacities of cyclic siloxanes (D3 to D8) with Gaussian 16 on UA supercomputers: CREST conformer searches, hindered-rotor frequencies, Boltzmann weighting.
- Built chee-research, a web app that pulls structures from PubChem, writes Gaussian inputs and SLURM scripts, and triages failed logs; cut days of prep per batch to one command; adopted by 3 research groups.
- Own the tooling for a 4-researcher, 1,500-species study testing whether one conformer gives engineering-grade heat capacities; wrote the duplicate-conformer detection method for the paper.
- Root-caused repeated hindered-rotor failures with Gaussian support (stale internal coordinates) and fixed Boltzmann-weighting bugs in the group’s analysis tool.

Otmow — São Paulo, Brazil

Product Management Intern, fintech (lending) — *July 2025 – Aug 2025*

- Wrote the PRD for a sales analytics dashboard through repeated 1:1s with the sales lead, turning a vague ask into scoped, prioritized requirements.
- Wrote a PRD with the head of finance for a lender credibility view; ran the build-vs-buy evaluation on VoIP tooling.

Traxda LLC — Tucson, AZ

Shop Foreman, promoted from Manufacturing Technician — *May 2023 – June 2025*

- Ran production for a suspension kit maker: laser and saw cutting, welding, powder coat, assembly, and shipping, about 370 part numbers.
- Set the daily build schedule, led a 4-person crew, trained technicians, and reached single-day order turnaround. Programmed Panasonic robotic welding cells from age 16.

SKILLS & ACTIVITIES

Product	PRDs, user interviews, clickable prototyping, pricing and competitive analysis, build-vs-buy
AI	Claude Code, multi-agent pipelines, MCP servers, LLM features in production, scikit-learn
Technical	Python (pandas, numpy), MATLAB, SQL/Postgres, TypeScript, REST APIs, Gaussian 16, HPC/SLURM, PI System
Interests	Product management, AI, startups, nuclear and other energy
Activities	FIRST Robotics fabrication lead (6-person subteam), Tucson Arizona Boys Chorus (7 years)